	(b)	(i)		Temp rise = $\Delta\theta$ = 640 [^{o}C] (1) Energy required to raise temp to boiling point = $mc\Delta\theta$ = 500 × 900 × 640 **ecf** = 288 000 000 [J] (1) Energy required to melt aluminium = 500 × 400 000 = 200 000 000 J (1) Total energy required = 4.88 × 10^8 J (1) **ecf** Substitution and conversion: time = $\frac{4.88 \times 10^8 \textbf{ ecf}}{8.7 \times 10^4}$ Answer **with unit** = 5 609.2 s **or** 93.5 min **or** 1.56 h (1) A conclusion consistent with their answer i.e. the one hour deadline is not met (1) **Alternative for 5th and 6th marks:** Number of kWh = 87 converted to J = 3.13 × 10^8 (1) This is less than the energy required **so** the one hour deadline is not met (1) **Alternative for 5th and 6th marks:** 4.88 × 10^8 J = 135.6 [kWh] (1) This is more energy than the heater can provide in one hour **so** the one hour deadline is not met (1) **Alternative:** Energy supplied = 87 000 × 3 600 (1) = 3.13 × 10^8 [J] (1) Temp rise = $\Delta\theta$ = 640 [^{o}C] (1) 3.13 × 10^8 = $m(c\Delta\theta + L)$ so 3.13 × 10^8 = m (900 × 640 + 400 000) (1) $m = \frac{3.13 \times 10^8}{976\,000}$ = 320.7 [kg] (1) A conclusion consistent with their answer i.e. the one hour deadline is not met (1)			6		6	
		(ii)		Energy is needed to raise the temperature of heater and container / some energy is lost [to the surroundings]	1			1		
				Question 6 total	**7**	**0**	**6**	**13**	**6**	**0**